Powdery Mildew Detection on Cotton Leaves Using Image Classification:

A Path to Integrated Pest Management (IPM) Optimization for Farmers

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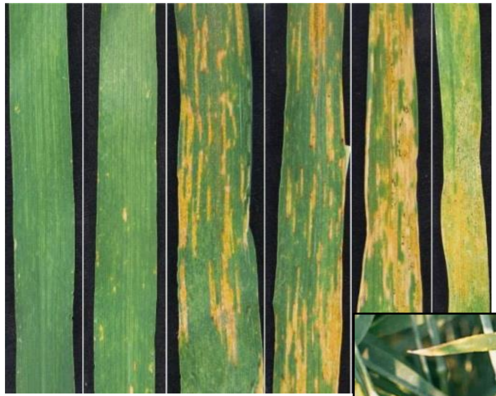
4/18/26

DSC 680 – T301



(Classifier model trained in detecting rust disease on wheat)

Issue:

Due to microclimate variations within crop fields, farmers often overuse resources such pesticides, fertilizers, and irrigation to compensate.

Precision agriculture push →

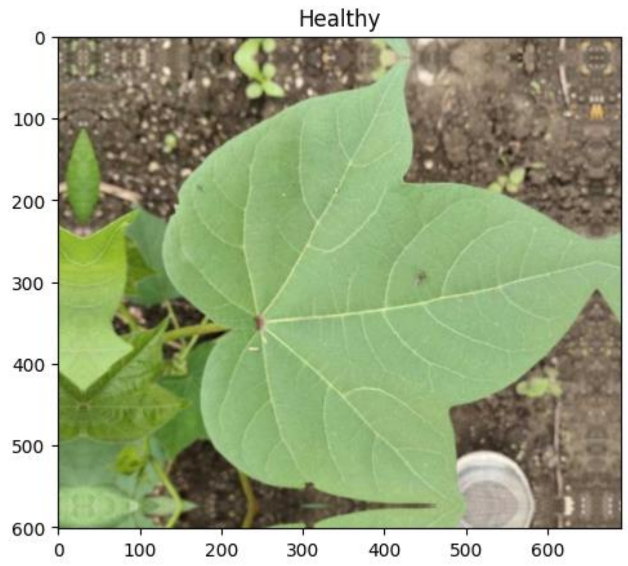
- Use technology to optimize resources
- Example:
  Combine robotics & image classification to locate pests/diseases faster for site-specific treatments
- Example pathogen:
  Powdery mildew, which can be treated with a fungicide

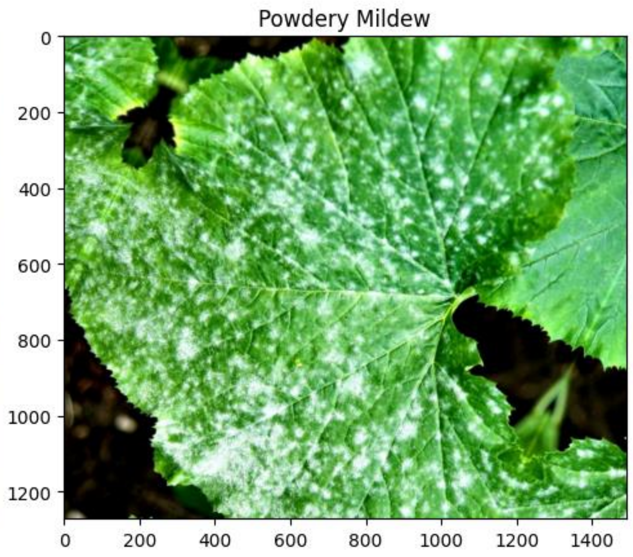
Project Goal:

Build a binary cotton-leaf image classification model that can identify healthy vs. diseased leaves

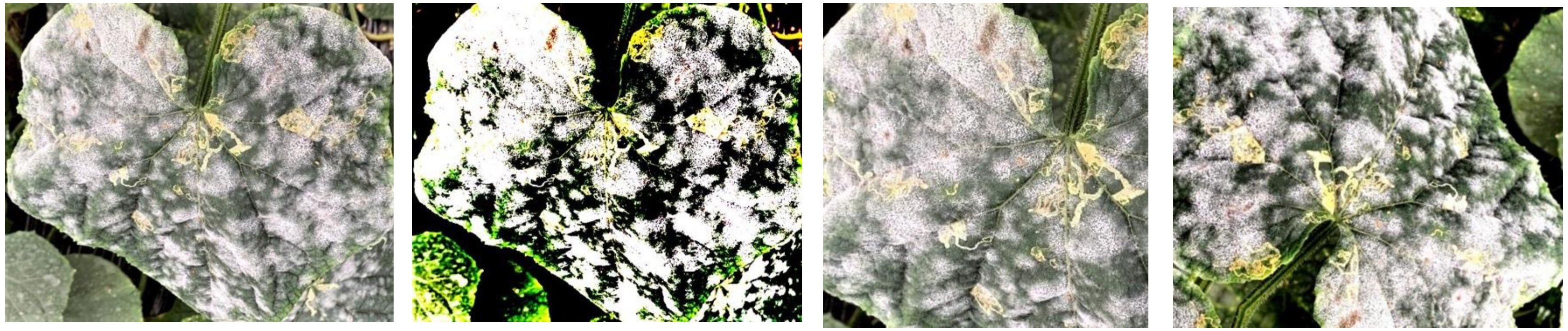
→ 1st step in developing a powdery mildew early warning system

Cotton-leaf Image Data





Augmented versions of the same image:

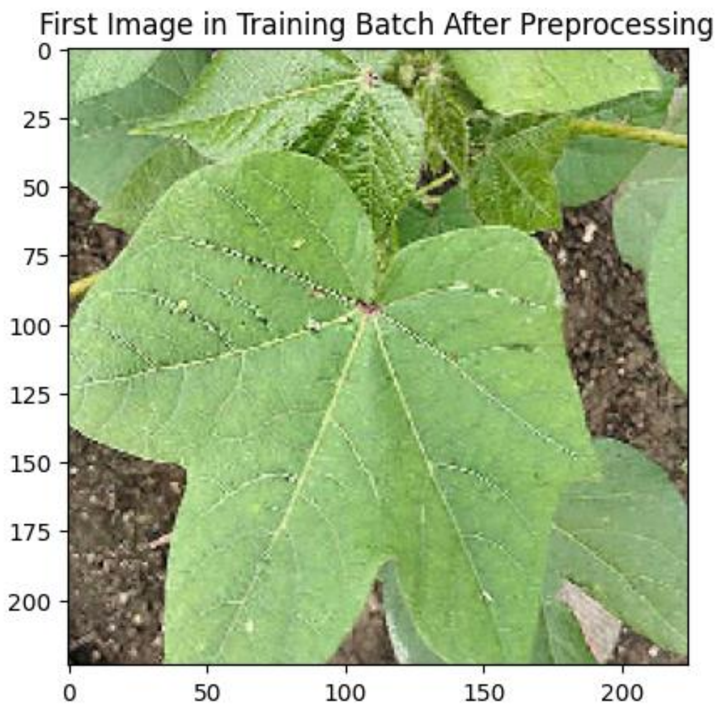


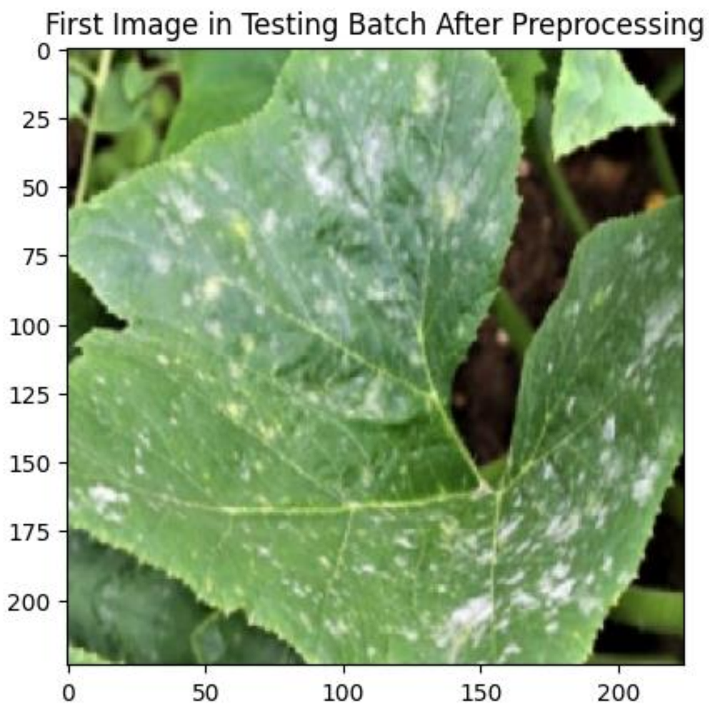
- Data set contains 800 images of **healthy** and **diseased** cotton leaves

- Data set includes **augmented versions** of original images
  Examples:
  - Rotated
  - Zoomed in
  - Adjusted brightness/contrast

- Augmented images **reduce model overfitting** by training to focus on recognizing patterns rather the specific colors/positioning

Data Preprocessing

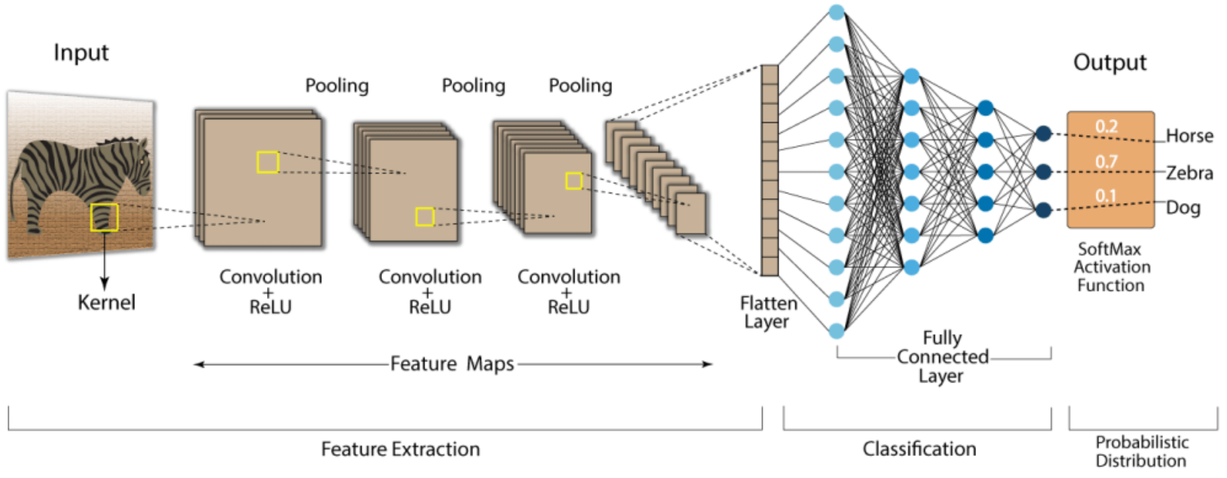




- Split cotton leaf data set into **80% training** and **20% testing**

- **To improve model processing efficiency:**
  - Normalize image data by dividing by 255 so that pixel values range between 0 →1
  - Resize images to standard height and width (224 x 224)
  - Set batch size to 32 so that model processes images in batches

Convolutional Neural Network (CNN)



Feature Extraction:

- **Convolutional layer →** use filters to detect feature patterns within pixel array; output is feature map
- **ReLU activation →** adds nonlinearity to features; helps model learn faster
- **Max pooling layer →** reduces feature map size to reduce processing load, overfitting and make model more tolerant to spatial variations

Flatten Layer → Converts 3D stacked feature maps into 1D vector

Classification:

- **Dense layer →** artificial neural network that acts as binary classifier
  - 0 = Healthy while 1 = Diseased
  - If predicted probability > 0.5, then image labeled diseased

CNN Model Building

Three different CNN model architectures were trained and evaluated:

Model #1:

- Convolutional layer & ReLU (Filters = 10)
- Convolutional layer & ReLU (Filters = 10)
- Convolutional layer & ReLU (Filters = 10)
- Flatten layer
- Dense (1 neuron)

Model #2:

- Convolutional layer & ReLU (Filters = 10)
- Max Pooling
- Convolutional layer & ReLU (Filters = 10)
- Max Pooling
- Convolutional layer & ReLU (Filters = 10)
- Max Pooling
- Flatten layer
- Dense (1 neuron)

Model #3:

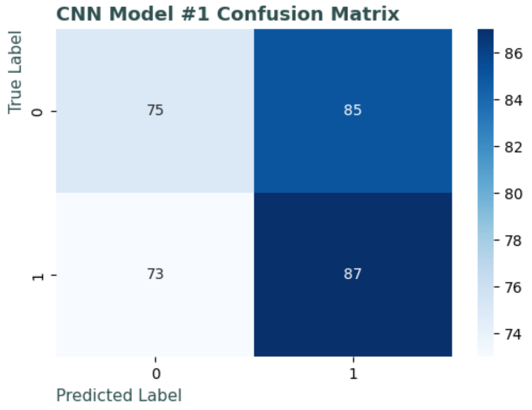
- Convolutional layer & ReLU (Filters = 32)
- Max Pooling
- Convolutional layer & ReLU (Filters = 64)
- Max Pooling
- Flatten layer
- Dense (64 neurons)
- Dense (1 neuron)

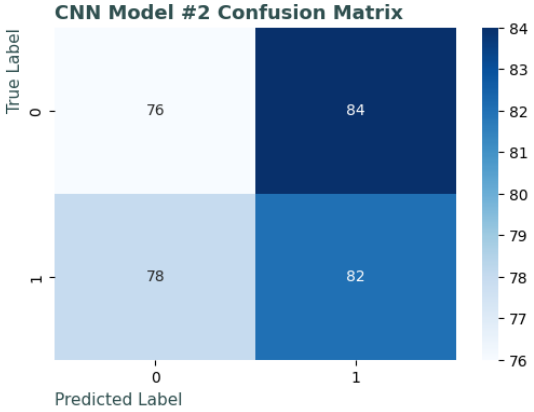
Model Evaluation

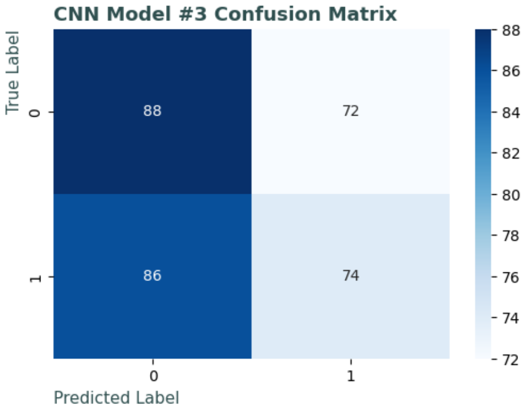
Overall, all three models performed similarly despite architecture variation.

- **1st model performed the worst** (higher binary cross-entropy and lower accuracy for validation) → likely due to lack of max pooling layers to help with spatial variation tolerance
- **Accuracy is high** (80% -99%) across all three models which may indicate overfitting

Model Evaluation Continued







Confusion matrices and calculated accuracy, precision, recall, and F1 score tell a different story.

- Calculated metrics using scikit-learn library are all roughly 50% for all three CNN models
- Sample distribution in confusion matrices roughly equal.

→→→→ Indicates models are no better at predicting correct classes than **random chance.**

Conclusions:

Would not recommend any of the current CNN models for deployment.

Ways to improve model:

- Investigate code for bugs to explain inconsistency between accuracy curve and scikit-learn metrics.
- Continue CNN architecture trial and error (ex. Increase number of filters).
- Fix data leakage between training and testing data sets.

If model is improved in powdery mildew detection, it could help target specific plants for treatment instead of field-wide pesticide application.

→ **benefits both the environment and farmer financial costs**

Questions

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